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Suspension assembly with slider arrangement

Abstract

A vehicle suspension arrangement includes a suspension slider assembly including first and second laterally spaced longitudinal frame members configured to be slidably coupled to a first vehicle frame member, first and second gusset members fixed to the first and second longitudinal frame members and each including a relief, a first lateral frame member extending between the first and second longitudinal frame members, the first lateral frame member having ends received within the reliefs, an axle member configured to support a pair of wheel assemblies, and a suspension assembly configured to support the suspension slider assembly from the axle member, the suspension assembly including a spring member positioned between the suspension slider assembly and the axle member.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This application is a continuation of U.S. patent application Ser. No. 17/814,633, filed Jul. 25, 2022, entitled “SUSPENSION ASSEMBLY WITH SLIDER ARRANGEMENT,” now U.S. Pat. No. 11,970,210, which claims the benefit of U.S. Provisional Patent Application No. 63/226,520, filed on Jul. 28, 2021, entitled “SUSPENSION ASSEMBLY WITH SLIDER ARRANGEMENT,” the entire disclosures of which are incorporated herein by reference.

BACKGROUND OF THE INVENTION

(1) The embodiments as shown and described herein relate to a vehicle suspension arrangement, particularly for heavy duty vehicles, and specifically a vehicle suspension arrangement that allows sliding repositioning of a supporting suspension assembly with respect to an associated vehicle frame assembly.

SUMMARY OF THE INVENTION

(2) One embodiment as disclosed herein may include a vehicle suspension arrangement that includes a suspension slider assembly including a first longitudinal frame member configured to be slidably coupled to a first vehicle frame member, a second longitudinal frame member laterally spaced from the first longitudinal frame member and configured to be slidably coupled to a first vehicle frame member, a first gusset member fixed to the first longitudinal frame member and including a first relief, a second gusset member fixed to the second longitudinal frame member and including a second relief, and a first lateral frame member extending between the first and second longitudinal frame members, the first lateral frame member having a first end received within the first relief and a second end received within the second relief, an axle member configured to support a pair of wheel assemblies, and a suspension assembly configured to support the suspension slider assembly from the axle member, the suspension assembly including a spring member positioned between the suspension slider assembly and the axle member.

(3) Another embodiment as disclosed herein may further or alternatively include a vehicle suspension arrangement that includes a suspension slider assembly configured to be slidably coupled to a vehicle frame assembly, an axle member configured to support a pair of wheel assemblies, and a suspension assembly configured to support the suspension slider assembly from the axle member, the suspension assembly including at least one leaf spring having a first end and a second end, a first hanger bracket pivotably coupling the first end of the at least one leaf spring to the suspension slider assembly, wherein the first hanger bracket is coupled to the suspension slider assembly via a first mechanical fastener, a second hanger bracket pivotably coupling the second end of the leaf spring to the suspension slider assembly, wherein the second hanger bracket is coupled to the suspension slider assembly via a second mechanical fastener, and a trailing arm having a first end pivotably coupled to the first hanger bracket, and a second end operably coupled to the axle member.

(4) Yet another embodiment as disclosed herein may further or alternatively include a vehicle suspension arrangement that includes a suspension slider assembly configured to be slidably coupled to a vehicle frame assembly, an axle member configured to support a pair of wheel assemblies, and a suspension assembly configured to support the suspension slider assembly from the axle member, the suspension assembly including a first leaf spring having a first end and a second end, a second leaf spring having a first end and a second end, a first hanger bracket operably coupled to the suspension slider assembly and pivotably coupling the first end of the first leaf spring to the suspension slider assembly, a second hanger bracket operably coupled to the suspension slider assembly and pivotably coupling the first end of the second leaf spring to the suspension slider assembly, and a center hanger bracket assembly that includes a third hanger bracket operably coupled to the suspension slider assembly, and an equalizing bracket pivotably coupled to the third hanger bracket and fixedly coupled to the second end of the first leaf spring and the second end of the second leaf spring.

(5) These and other features, advantages, and objects of the present invention will be further understood and appreciated by those skilled in the art by reference to the following specification, claims, and appended drawings.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) FIG. 1 is a perspective view of a vehicle suspension arrangement;
- (2) FIG. 2 is a perspective view of a suspension assembly;
- (3) FIG. 3 is a perspective view of a suspension slider assembly;
- (4) FIG. 4 is an exploded perspective view of the suspension slider assembly;
- (5) FIG. 5 is an exploded perspective view of a vehicle frame rail and a gusset;
- (6) FIG. 5A is a perspective view of an alternative embodiment of the gusset;
- (7) FIG. 6 is a perspective view of the gusset;
- (8) FIG. 7 is a perspective view of an alternative embodiment of the suspension slider assembly;
- (9) FIG. 8 is an exploded perspective view of the alternative embodiment of the suspension slider assembly;
- (10) FIG. 9 is a partial perspective view of the alternative embodiment of the suspension slider assembly;
- (11) FIG. 10 is a perspective view of a gusset of the suspension slider assembly of FIG. 7;
- (12) FIG. 11 is an exploded perspective view of another alternative embodiment of the suspension slider assembly;
- (13) FIG. 12 is a partial perspective view of the suspension slider assembly of FIG. 11;
- (14) FIG. 12A is an enlarged perspective view of the area XIIA, FIG. 12;
- (15) FIG. 13 is a partial perspective view of the suspension slider assembly of FIG. 11;
- (16) FIG. 14 is a perspective view of a gusset of the suspension slider assembly of FIG. 11;
- (17) FIG. 14A is a partial perspective view of another alternative embodiment of the suspension slider assembly;
- (18) FIG. 15 is an exploded perspective view of the vehicle frame rail, a first hanger bracket, and a wear plate mounting bracket;
- (19) FIG. 16 is a cross-sectional side elevation view taken along the line XVI-XVI, FIG. 1;
- (20) FIG. 17 is an exploded perspective view of the vehicle frame rail, a second hanger bracket, and an equalizing bracket;
- (21) FIG. 18 is a cross-sectional side elevation view taken along the line XVIII-XVIII, FIG. 1;
- (22) FIG. 19 is a cross-sectional side elevation view taken along the line XIX-XIX, FIG. 1;
- (23) FIG. 20 is a cross-sectional side elevation view taken along the line XX-XX, FIG. 1;
- (24) FIG. 21 is a perspective view of an actuator cam; and
- (25) FIG. 22 is a perspective view of an alternative actuator cam arrangement.

DETAILED DESCRIPTION OF THE PREFERRED EMBODIMENTS

- (26) For purposes of description herein, the terms “upper,” “lower,” “right,” “left,” “rear,” “front,” “vertical,” “horizontal,” and derivatives thereof shall relate to the invention as oriented in FIG. 1. However, it is to be understood that the invention may assume various alternative orientations, except where expressly specified to the contrary. It is also to be understood that the specific devices and processes illustrated in the attached drawings, and described in the following specification are simply exemplary embodiments of the inventive concepts defined in the appended claims. Hence, specific dimensions and other characteristics relating to the embodiments disclosed herein are not to be considered as limiting, unless the claims expressly state otherwise.
- (27) As best illustrated in FIG. 1, a vehicle suspension arrangement 10 slidably engages a vehicle frame assembly 12 to support an associated overall vehicle assembly at various preselected

positions along a length of the vehicle frame assembly **12**. In the illustrated example, the vehicle suspension arrangement **10** includes a suspension slider assembly **14** that slidably engages the vehicle frame assembly **12**, a pair of axle members including a front axle member **16** and a rear axle member **18**, where the overall vehicle assembly travels in a direction **20** and the axle members **16**, **18** support wheel assemblies **22**, and a suspension assembly **24** operably supporting the suspension slider assembly **14** from the axle members **16**, **18**.

(28) The vehicle frame assembly **12** may include a pair of longitudinally-extending vehicle frame rails **26** each having a Z-shaped cross-sectional configuration and a plurality of locking apertures **28** spaced along the length thereof.

(29) The suspension assembly **24** (FIG. 2) may include a plurality of leaf spring assemblies including a pair of front leaf spring assemblies **32** and a pair of rear leaf spring assemblies **34**. In the illustrated example, the front leaf spring assemblies **32** are operably coupled to the suspension slider assembly **14** via respective front hanger brackets **36** and center hanger bracket assemblies **38**, while the rear leaf spring assemblies **34** are operably coupled to the suspension slider assembly **14** via respective rear hanger brackets **40** and the center hanger bracket assemblies **38**, as further described below. Although, the present example of the suspension assembly **24** described herein includes leaf spring assemblies **32**, **34**, other spring elements may also be utilized including, but not limited to, air spring assemblies depending on the overall application.

(30) In one embodiment, the suspension slider assembly **14** (FIGS. 3 and 4) may include a pair of laterally spaced longitudinal frame members **42** each having a vertically-extending body portion **43**, an upper flange **44** extending orthogonally inward from the body portion **43**, and a lower flange **46** extending substantially orthogonally inward from the body portion **43**. The suspension slider assembly **14** may further include a plurality of box-shaped lateral frame members, including a front lateral frame member **48**, a rear lateral frame member **50** and a center lateral frame member **52**, where the lateral frame members **48**, **50**, **52** are longitudinally spaced from one another along the length of the longitudinal frame members **42**. In the illustrated example, each of the lateral frame members **48**, **50**, **52** have a box-shaped cross-sectional configuration, where the front lateral frame member **48** has a downwardly and rearwardly opening, longitudinally-extending slot **54**, the rear lateral frame member **50** includes a downwardly and forwardly opening, longitudinally-extending slot **56**, and the center lateral frame member **52** includes a downwardly, longitudinally-extending slot **58**. The lateral frame members **48**, **50**, **52** may be provided with alternative cross-sectional configurations, as further described below. The suspension slider assembly **40** further includes a plurality of gusset members configured to attach the lateral frame members **48**, **50**, **52** to the longitudinal frame members **42**, including a pair of front gusset members **60**, a pair of rear gusset members **62** similarly configured as the front gusset members **60**, and a pair of center gusset members **64**. By way of example, one of the front gusset members **60** is described below, and it is noted that the front gusset members **60** and the rear gusset members **62** are similarly constructed. In the illustrated example, each front gusset members **60** (FIGS. 5 and 6) includes a body portion **66** having an aperture **68** extending therethrough, and a pair of side walls **70** each extending orthogonally outward from the body portion **66**. Each side wall **70** may include an alignment tab **72** located along the vertical length thereof, and a recessed area **74** that includes a pair of apertures **76** configured to receive mechanical fasteners therethrough for fastening the front gusset member **60** to the front lateral frame member **48**, as described below. An alternatively-configured gusset **60a** (FIG. 5A) may include an over-bent or recessed sidewalls **70a** rather than a recessed area **74** as described above with respect to the gusset **60**. Since the gusset **60a** is similar to the gusset **60**, similar parts or elements appearing in FIG. 5 and FIG. 5A respectively are represented by the same, corresponding reference numeral, except for the suffix “a” in the numerals of the latter. During assembly, the alignment tabs **72** are positioned within corresponding alignment slots **78** extending through the body portion **43** of the associated longitudinal frame member **42**, thereby assuring proper alignment of the front gusset member **60** with the longitudinal frame member **42**. A weld **80**

(FIG. 1) is then placed about the periphery of the front gusset member **60**, thereby fixedly securing the front gusset member **60** with the longitudinal frame member **42**. The rear gusset members **62** and the center gusset members **64** include similar alignment features, such as tabs similar to those as discussed above with respect to the front gusset member **60** and that cooperate with corresponding and appropriately located slots within the longitudinal frame member **42**, where each gusset member **62**, **64** is weldably secured to the associated longitudinal frame member **42** during assembly.

(31) As best illustrated in FIGS. 3 and 4, the ends **82** of the lateral frame members **48**, **50**, **52** are inserted into the aperture **68** of the gusset members **60**, **62**, **64** and are secured thereto via a plurality of mechanical fasteners, such as bolts **84** that extend through the apertures **76** (FIGS. 5 and 6) and through corresponding apertures **86**, **68** located at the ends **82** of the lateral frame members **48**, **50**, **52** and are held in place by corresponding nuts (not shown). It is noted that the non-welded, modular construction of the suspension slider assembly **14** provided via the mechanical fastening arrangements allow portions of the suspension slider arrangement **14** to be prefabricated, and the final assembly to be done on-site or at a facility where an associated trailer assembly is being constructed, and allows the installer/manufacturer, to easily adjust the overall width of the suspension slider assembly **14** to fit vehicles having various spacing between the longitudinal frame members **42**. It is noted that other mechanical fasteners configured to provide such advantages may also or alternatively be utilized.

(32) The reference numeral **14a** (FIGS. 7 and 8) generally designates another embodiment of the suspension slider assembly. Since the suspension slider assembly **14a** is similar to the previously described suspension slider assembly **14**, similar parts appearing in FIGS. 1-6 and FIGS. 7 and 8 respectively are represented by the same, corresponding reference numeral, except for the suffix "a" in the numerals of the latter. In the illustrated example, the front lateral frame member **48a**, the rearward lateral frame member **50a** and the center lateral frame member **52a** are each provided with a C-shaped cross-sectional configuration, with a plurality of weight-reducing apertures **88** that may be spaced along the length thereof. Each lateral frame member **48a**, **50a**, **52a** includes a body portion **90**, and an upper flange **92** and a lower flange **94** each extending orthogonally from the body portion **90**. The suspension slider assembly **14a** may further include a pair of front gusset members **60a**, a pair of rear gusset members **62a** and a pair of center gusset members **64a**. As the gusset members **60a**, **62a**, **64a** may be similarly configured, by way of example each rear gusset member **62a** (FIGS. 9 and 10) includes a body portion **96**, and a side wall **98** extending orthogonally from the body portion **96**. The body portion **96** may include a weight-reducing aperture **100** extending therethrough. The body portion **96** may further include a pair of alignment tabs **102** while the front lateral frame member **50a** may include a pair of corresponding alignment slots **104** that cooperate to assist with proper alignment of the front lateral frame member **48a** with the corresponding longitudinal frame member **42a** during assembly. The body portion **96** may further include cutouts **106** that provide spacing for the upper and lower flanges **92**, **94** of the front lateral frame member **48a** between the front gusset member **60a** and the upper and lower flanges **44a**, **46a** of the associated longitudinal frame member **42a**. As best illustrated in FIG. 9, a weld **108** may be placed about the periphery of the front gusset member **68** subsequent to proper alignment of the front lateral frame member **48a** with the front gusset member **60a** and the longitudinal frame member **42a**, thereby fixedly securing the front lateral frame member **48a** with the longitudinal frame member **42a**.

(33) The reference numeral **14b** (FIGS. 11 and 12) generally designates another embodiment of the suspension slider assembly. Since the suspension slider assembly **14b** is similar to the previously described suspension slider assembly **14**, similar parts appearing in FIGS. 1-6 and FIGS. 11 and 12 respectively are represented by the same, corresponding reference numeral, except for the suffix "b" in the numerals of the latter. In the illustrated example, the suspension slider assembly **14b** includes a front lateral frame member **48b**, a rear lateral frame member **50b** and a center lateral

frame member **52b**, each having a Z-shaped cross-sectional configuration. As each of the lateral frame members **48b**, **50b**, **52b** are similarly-configured, the rear lateral frame member **50b** is described herein. In the illustrated example, each rear lateral frame member **50b** includes a body portion **110** with a plurality of weight-reducing apertures **112** that may be spaced along the length thereof, an upper flange **114** extending orthogonally from the body portion **110**, and a lower flange **116** extending orthogonally from the body portion **110** and opposite the upper flange **114**. It is noted that the rearwardly-extending lower flange **116** (FIGS. **12** and **12A**) may be utilized to support accessories and other components from the suspension slider assembly **14a**, such as mud flaps, aerodynamic improvement components, and the like.

(34) The suspension slider assembly **14b** may further include a pair of front gusset members **60b**, a pair of rear gusset members **62b** and a pair of center gusset members **64b**. As the gusset members **60b**, **62b**, **64b** are similarly configured, the front gusset member **60b** (FIGS. **13** and **14**) is described herein. In the illustrated example, each front gusset member **60b** includes a body portion **118**, and a pair of side walls **120** extending orthogonally from the body portion **118**. In the illustrated example, the body portion **118** may include a weight reduction aperture **122** extending therethrough.

Alternatively, the weight reduction aperture **122** may be configured so as to receive ends of a lateral frame member, similar to the previously described arrangement of the lateral frame members **48**, **50**, **52** with respect to the gussets **60**, **62**, **64** of the suspension slider assembly **14**, thereby allowing the front gusset members **60b**, **62b**, **64b** to alternatively be used in conjunction with the Z-shaped lateral members **48b**, **50b**, **52b**, or the box-shaped lateral frame members **48**, **50**, **52**. The upper edge of the body portion **118** and one of the side walls **120** may include a cutout **122** that provides space for positioning the upper flange **114** of the front lateral frame member **48b** between the front gusset member **60b** and the upper flange **44b** of the longitudinal frame member **42b**. The front gusset member **60b** may be fixedly secured to the longitudinal frame member **42b** and/or the front lateral frame member **48** by welds **123** extending about the periphery thereof. Alternatively, the front gusset member **60b** may be weldably secured to the associated longitudinal frame member **42b**, while the front lateral frame member **48b** is secured to the front gusset member **60b** via a plurality of mechanical fasteners, such as bolts **124** and **126** as best illustrated in FIGS. **11** and **12**.

(35) The reference numeral **14c** (FIG. **14A**) generally designates another embodiment of the suspension slider assembly. Since the suspension slider assembly **10c** is similar to the previously described suspension slider assembly **14**, similar parts appearing in FIGS. **1-6** and FIG. **14A** respectively are represented by the same, corresponding reference numeral, except for the suffix "c" in the numerals of the latter. In the illustrated example, at least one of the lateral frame members has an I-beam shaped cross-sectional configuration, where FIG. **14A** shows the center lateral frame member **52c** as an example. The center lateral frame member **52c** may have a vertical body portion **350**, an upper flange **352** extending orthogonally from the vertical body portion **350**, and a lower flange **354** extending orthogonally from the vertical body portion **350**. The center lateral frame member **52c** is releasably secured to the longitudinal frame members **42c** via a plurality of mechanical fasteners, and specifically may be secured to the longitudinal frame members **42c** by a plurality of bolts **356** that respectively secure the upper and lower flanges **352**, **354** of the center lateral frame member **52c** to the upper and lower flanges **44c**, **46c** of the longitudinal frame members **42c**. It is noted that this connection arrangement may also be utilized with and incorporated for use with beams with various other configurations, including but not limited to, the box-shaped lateral frame members **48**, **50**, **52** (FIGS. **1**, **3** and **4**), the C-shaped lateral frame members **48a**, **50a**, **52a** (FIGS. **7** and **8**), and the Z-shaped lateral frame members **48b**, **50b**, **52b** (FIGS. **11** and **12**).

(36) As described above, the suspension assembly **24** (FIGS. **1** and **2**) includes the front leaf spring assemblies **32**, the rear leaf spring assemblies **34**, the front hanger brackets **36**, the center hanger bracket assemblies **38** and the rear hanger brackets **40**. As the front hanger brackets **36** and the rear

hanger brackets **40** are similarly constructed, only the front hanger brackets **36** are described herein. As best illustrated in FIGS. **15** and **16**, each front hanger bracket **36** has a U-shaped configuration and includes an upper wall **128** and a pair of side walls **130** extending orthogonally downward from the upper wall **128**. The top wall includes a pair of alignment apertures configured to align with a corresponding pair of alignment apertures **138** extending through the bottom flange **46** of the longitudinal frame member **42**. Each side wall **130** further includes an integrally formed stop member **140**, and a pair of apertures **142** located at a distal end thereof and configured to pivotally mount a trailing arm, as described below. A wear pad assembly **144** includes a wear pad mounting bracket **146** and a wear pad **148** secured to an outer surface of the wear pad mounting bracket **146**. The wear pad mounting bracket **146** may include a pair of alignment tabs **150** extending therefrom. In assembly, the wear pad assembly **144** is positioned within the associated front hanger bracket **36** such that the alignment tabs **150** align with and extend through the alignment apertures **136** of the front hanger bracket **36**. The front hanger bracket **36** and the wear pad assembly **144** are then positioned and aligned with the longitudinal frame member **42** such that the alignment tabs **50** are aligned with and received within the corresponding alignment apertures **138** of the longitudinal frame member **142**, thereby properly aligning the front hanger bracket **36** and the wear pad assembly **144** with the longitudinal frame member **42**. The wear pad mounting bracket **146** and the front hanger bracket **36** may then be secured to the bottom flange **46** of the longitudinal frame member **42** via welds **152**. Alternatively or in addition, the wear pad mounting bracket **146** may be supported via bracket support portions **154** that extend inwardly from the side walls **130** of the front hanger bracket **36**.

(37) As best illustrated in FIGS. **17** and **18**, each center hanger bracket assembly **38** includes a center hanger bracket **156** fixedly attached to the longitudinal frame member **42** and an equalizer bracket **158** pivotably coupled to the center hanger bracket **156**. In the illustrated example, the center hanger bracket **156** has a U-shaped cross-sectional configuration and includes an upper wall **160** and a pair of side walls **162** extending orthogonally downward from the upper wall **160**. The upper wall **160** may include a pair of stop members **164** and bosses or extensions **166**, as further described below. Each side wall **162** includes an upper aperture **168** for pivotably coupling the equalizer bracket **158** with the center hanger bracket **156** and a lower aperture **174** pivotably coupling the center hanger bracket **156** with a corresponding trailing arm, as described below. The equalizer bracket **158** includes an arcuately-shaped upper wall **172** and a pair of side walls **174** extending orthogonally downward from the upper wall **172** and weldably secured therewith. The side walls **174** may each include a pair of integrally formed stop members **176** and a centrally located pivot aperture **178**. In assembly, the alignment bosses or extensions **166** are aligned with and positioned within corresponding alignment apertures **180** of the lower flange **46** of the longitudinal frame member **42**, thereby ensuring proper alignment of the center hanger bracket **156** with the longitudinal frame member **42** before the center hanger bracket **156** is fixedly secured to the longitudinal frame member **42** via a weld **182**. The equalizer bracket **158** is pivotably secured to the center hanger bracket **156** via a pivot assembly or bushing assembly **184**, where the stop members **164** are configured to limit pivoting movement of the equalizer bracket **158** with respect to the center hanger bracket **156** during operation of the vehicle suspension arrangement **10**.

(38) As best illustrated in FIG. **19**, each front leaf spring assembly **32** includes a center portion **186** secured to the front axle member **16**, a first end **188** positioned between the wear pad assembly **144** and the stop member **140** of the front hanger bracket **36**, and a second end **190** positioned between the equalizer bracket **158** and a pair of the stop members **176**. Similarly, the rear leaf spring assemblies **34** each include a center portion **192** secured to the rear axle member **18**, a first end **174** positioned between the wear pad assembly **144** and the stop members **140**, and a second end **196** positioned between the equalizer bracket **158** and stop members **176**. It is noted that wear pads **198** may be positioned between the top upper wall **160** of the equalizer bracket **158** and the second ends **190**, **196** of the front and rear leaf spring assemblies **32**, **34**. A pair of forward trailing arms **200**

extend between the corresponding front hanger bracket **36** and the front axle member **16**, with a first end **202** of each forward trailing arm **200** pivotably secured to a corresponding front hanger bracket **36** at the apertures **142** via a pivot or bushing assembly **204**, and with a second end **206** of each of the trailing arms **200** pivotably secured to the front axle member **16** at a pivot point **208** via a corresponding pivot or bushing assembly **210**. Similarly, a pair of rearward trailing arms **212** pivotably couple the center hanger bracket **156** with the rear axle member **18**, with each rearward trailing arm **212** including a first end **214** pivotably coupled to the corresponding center hanger bracket **156** at the lower apertures **170** via a pivot or bushing assembly **216** and a second end **218** of the trailing arm **212** pivotably coupled to the rear axle member **18** via a pivot or bushing assembly **220** for pivoting about a pivot point **222**.

(39) In assembly, the suspension slider assembly **14** is slidably coupled to the vehicle frame assembly **12** via a plurality of slider brackets **223** secured to the body portion **43** of the longitudinal frame members **42** and each including a U-shaped section that extends about an outwardly-extending lower flange **224** of the corresponding Z-shaped vehicle frame rail **26**, thereby allowing the suspension slider assembly **14** to be slidably repositioned along the length of the vehicle frame assembly **12**. A friction reduction member (**227**) may be located between an upper flange **225** of each of the vehicle frame rails **23** and the corresponding longitudinal frame member **42** of the suspension slider assembly **14**.

(40) The vehicle suspension arrangement **10** may further include a locking arrangement **220** (FIGS. **1**, **19** and **20**) that includes a longitudinally-extending torsional bar **226** having a first end **228** pivotably received within the front lateral frame member **48** and a second end **230** pivotably received within the rear lateral frame member **50**, and a biasing member comprising a coil spring **231** operably coupled to manual input lever or handle **233**. In the illustrated example, a center portion **232** of the torsional bar **226** extends over an upper wall **234** of the center lateral frame member **52** and is secured thereto via a pivot bracket **236**. Alternatively, and as best illustrated in FIGS. **7**, **8** and **14a**, the center lateral frame member, including the center lateral frame members **52**, **52a**, **52b**, may be configured such that the torsional bar **226a**, **226c** extends through the center lateral frame member **52a**, **52c** rather than passing over or above it, and that a similar configuration could be utilized with the suspension slider assembly **14b** (FIGS. **11** and **12**). The torsional bar **226** is coupled to corresponding locking pins **240** via corresponding actuator arms **242** pivotably coupled to the torsional bar **226** via actuator cams **244**. As best illustrated in FIG. **21**, the actuator cams **244** each include a body portion **301** having an aperture **303** extending therethrough which receives the torsional bar **226**, and a pair of actuator arm apertures **305** for receiving ends of the respective actuator arms **242** (FIG. **1**), where the body portion is welded to the torsional bar **36**. In an alternative embodiment, a bolt-on actuator cam arrangement **307** (FIG. **22**) may be provided, where the actuator cam arrangement **307** includes a pair of collar members **309** each having a cylindrical body portion **311** clamped to one another about the torsional bar **226** via a pair of clamping bolts **313**, and an arm portion **315** extending outward from the respective body portion **311** and pivotably coupled to an end of a respective actuator arm **242**. Each locking pin **240** is slidably received within a locking pin mounting bracket **246** that is fixedly secured to the body portion **43** of the corresponding longitudinal frame member **42** via a weld. It is noted that in the illustrated example, each locking pin mounting bracket **246** includes a pair of apertures for receiving the locking pins **240**, including a small diameter aperture **248** and a large diameter aperture **250**, where the large diameter aperture **250** has a larger diameter than the small diameter aperture **248**. It is noted that this configuration of the locking pin mounting bracket **246** allows a standardized, single assembly to be utilized with locking pins of various diameters, where the particular locking pin may be selected to correspond to the diameter of the locking apertures **28** of the vehicle frame rails **26**. In operation, the locking arrangement **220** is operable between a locked configuration, where the locking pins **240** extend through the locking apertures **28** of the vehicle frame rails **26**, thereby preventing sliding movement of the suspension slider assembly **14** with

respect to the vehicle frame assembly **12**, and an unlocked configuration where the locking pins **40** are withdrawn from the locking apertures **28** of the vehicle frame rails **26**, thereby allowing the suspension slider assembly **14** to be slidably repositioned along the length of the vehicle frame assembly **12**, where the locking arrangement is biased toward the locking configuration by the biasing member **231**.

Claims

1. A vehicle suspension arrangement, comprising: a suspension slider assembly, comprising: a first longitudinal frame member configured to be slidably coupled to a first vehicle frame member; a second longitudinal frame member configured to be slidably coupled to a second vehicle frame member, wherein at least one of the first vehicle frame member and the second vehicle frame member includes a plurality of locking apertures spaced along a length thereof; a locking arrangement, comprising: a locking member connected to the first longitudinal frame member and including a first locking member receiving aperture having a first diameter that is substantially similar to a diameter of each of the plurality of locking apertures and a second locking member receiving aperture having a second diameter that is different than the diameter of the first locking member receiving aperture, wherein the first diameter is less than the second diameter; and a locking pin movable between an engaged position where the locking pin engages the first locking member receiving aperture and one of the plurality of locking apertures such that the suspension slider assembly is prevented from sliding with respect to the first and second vehicle frame members, and a disengaged position where the locking pin is removed from engagement with the one of the plurality of locking apertures such that the suspension slider assembly is slidably repositionable with respect to the first and second vehicle frame members; an axle member configured to support a pair of wheel assemblies; and a suspension assembly configured to support the suspension slider assembly from the axle member, the suspension assembly includes a spring member positioned between the suspension slider assembly and the axle member.
2. The vehicle suspension arrangement of claim 1, wherein the locking member is a single, integral piece.
3. The vehicle suspension arrangement of claim 1, wherein the locking member is one of a plurality of locking members spaced along a length of at least one of the first and second longitudinal frame members.
4. The vehicle suspension arrangement of claim 3, where a distance between the first locking member receiving aperture and the second locking member receiving aperture of each locking member is less than a distance between a pair of adjacent locking members of the plurality of locking members.
5. The vehicle suspension arrangement of claim 1, wherein the locking arrangement further comprises a torsion bar operably coupled to the locking pin via a linking member.
6. The vehicle suspension arrangement of claim 5, where the torsion bar is coupled to the linking member via a cam member.
7. The vehicle suspension arrangement of claim 5, wherein the torsion bar is configured to be rotated via a manual input lever.
8. The vehicle suspension arrangement of claim 1, wherein the locking arrangement further comprises a biasing member configured to bias the locking pin from the disengaged position toward the engaged position.
9. The vehicle suspension arrangement of claim 1, where a distance between the first locking member receiving aperture and the second locking member receiving aperture of the locking member is less than a distance between a pair of adjacent locking members of the plurality of locking members.
10. The vehicle suspension arrangement of claim 1, wherein the spring member comprises a leaf

spring.

11. A vehicle suspension arrangement, comprising: a suspension slider assembly, comprising: a first longitudinal frame member configured to be slidably coupled to a first vehicle frame member; a second longitudinal frame member configured to be slidably coupled to a second vehicle frame member, wherein at least one of the first vehicle frame member and the second vehicle frame member includes a plurality of locking apertures spaced along a length thereof; and a locking arrangement, comprising: a plurality of locking members connected to and spaced along the first and second longitudinal frame members and each including a first locking member receiving aperture having a first diameter that is substantially similar to a diameter of each of the plurality of locking apertures and a second locking member receiving aperture having a second diameter that is different than the diameter of the first locking member receiving aperture, wherein the first diameter is less than the second diameter; a plurality of locking pins spaced along the first and second longitudinal frame members and each movable between an engaged position where the plurality of locking pins engage the first locking member receiving aperture of a respective one of the plurality of locking members and a respective one of the plurality of locking apertures such that the suspension slider assembly is prevented from sliding with respect to the first and second vehicle frame members, and a disengaged position where the plurality of locking pins are removed from engagement with the respective one of the plurality of locking apertures such that the suspension slider assembly is slidably repositionable with respect to the first and second vehicle frame members; a biasing member configured to bias the plurality of locking pins from the disengaged position toward the engaged position; and a torsion bar operably coupled to the plurality of locking pins via a plurality of linking members and the plurality of linking members are coupled to the torsion bar via a plurality of cam members; an axle member configured to support a pair of wheel assemblies; and a suspension assembly configured to support the suspension slider assembly from the axle member, the suspension assembly include a spring member positioned between the suspension slider assembly and the axle member.

12. The vehicle suspension arrangement of claim 11, wherein each of the plurality of the locking members are a single, integral piece.

13. The vehicle suspension arrangement of claim 11, wherein the torsion bar is rotated via a manual input lever.

14. The vehicle suspension arrangement of claim 11, where a distance between the first locking member receiving aperture and the second locking member receiving aperture of each of the plurality of locking members is less than a distance between a pair of adjacent locking members of the plurality of locking members.

15. The vehicle suspension arrangement of claim 11, wherein the spring member comprises a leaf spring.
